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United States Patent	12402625
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Fabri; Carlos Eduardo et al.

Fungicidal combinations

Abstract

A combination including a dithiocarbamate fungicide, a succinate dehydrogenase fungicide and at least one of ergosterol biosynthesis inhibitor fungicide or a quinone outside inhibitor fungicide.

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Appl. No.: 17/130193

Filed: December 22, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20210106006 A1	Apr. 15, 2021

Foreign Application Priority Data

IN	201631037704	Nov. 04, 2016
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Related U.S. Application Data

continuation parent-doc US 16347283 PENDING WO PCT/IB2017/056710 20171030 child-doc US 17130193

Publication Classification

Int. Cl.: **A01N47/14** (20060101); **A01N37/36** (20060101); **A01N43/16** (20060101); **A01N43/40** (20060101); **A01N43/54** (20060101); **A01N43/56** (20060101); **A01N43/60** (20060101); **A01N43/653** (20060101); **A01N45/02** (20060101); **A01P3/00** (20060101); **A61F2/844** (20130101); **A61F2/92** (20130101); **A61F2/95** (20130101)

U.S. Cl.:

CPC **A01N47/14** (20130101); **A01N37/36** (20130101); **A01N43/16** (20130101); **A01N43/40** (20130101); **A01N43/54** (20130101); **A01N43/56** (20130101); **A01N43/60** (20130101); **A01N43/653** (20130101); **A01N45/02** (20130101); **A61F2/844** (20130101); **A61F2/92** (20130101); **A61F2/9522** (20200501); **A01N47/14** (20130101); **A01N43/40** (20130101); **A01N43/54** (20130101); **A01N43/56** (20130101); **A01N43/653** (20130101); **A01N45/02** (20130101); **A01N43/40** (20130101); **A01N43/54** (20130101); **A01N43/653** (20130101); **A01N43/653** (20130101); **A01N43/54** (20130101); **A01N45/02** (20130101); **A01N43/56** (20130101); **A01N43/653** (20130101); **A01N43/56** (20130101); **A01N43/54** (20130101); **A01N43/653** (20130101);

Field of Classification Search

CPC: A01N (47/14); A01N (37/36); A01N (43/60); A01N (43/653); A01N (43/56); A01N (43/54); A01N (43/40); A01G (13/00); A01P (3/00)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. application Ser. No. 16/347,283, filed on May 3, 2019, which is a National Stage application of PCT/IB2017/056710, filed on Oct. 30, 2017, which claims the benefit of Indian Application No. 201631037704, filed on Nov. 4, 2016, both of which are incorporated by reference in their entirety herein.

TECHNICAL FIELD

(1) The present invention relates to a combination of fungicides. More specifically, the present invention relates to fungicidal combinations comprising succinate dehydrogenase inhibitor fungicides for controlling a broad spectrum of fungal diseases.

BACKGROUND OF THE INVENTION

(2) Fungicides are an integral and important tool yielded by farmers to control diseases, as well as to improve yields and quality of the crops. There are various fungicides that have been developed over the years with many desirable attributes such as specificity, systemicity, curative and eradant action and high activity at low use rates.

(3) Succinate dehydrogenase inhibitor (SDHI) fungicides are known in the art to be broad spectrum and have a high potency. Pyrazolecarboxamides are a group of active compounds within the SDHI family of fungicides that are known to be more potent than most other SDHI fungicides. These molecules specifically bind to the ubiquinone-binding site (Q-site) of the mitochondrial complex II, thereby inhibiting fungal respiration. These fungicides are known to control a broad spectrum of fungal diseases.

(4) Various other classes of fungicides are also known in the art, such as Quinone outside inhibitors (Qols), ergosterol-biosynthesis inhibitors, fungicides that act on multiple sites, fungicides that affect mitosis etc. These fungicides have been mixed with SDHI fungicides to achieve a broad spectrum of disease control.

(5) WO2006037632 teaches combinations of SDHI fungicides with a second active compound.

WO2013127818 teaches combinations of SDHI fungicides with various herbicides.

WO2006037634 teaches methods of controlling fungi using a combination of SDHI fungicide with various fungicides. However, the prior art does not teach the use of ternary or higher combinations of SDHI fungicides.

(6) Dithiocarbamate are known in the art as multi-site fungicides. These fungicides are used for broad-spectrum disease control in more than 70 crops. Mancozeb is especially important for controlling devastating and fast spreading diseases such as *Phytophthora infestans*, *Venturia inaequalis* etc. Dithiocarbamate fungicides, especially mancozeb, are particularly useful for disease control because of their broad spectrum of activity, high tolerance by crop plants, and general

usefulness for controlling fungal plant diseases not controlled by active compounds that act on only a single target site in the fungus.

(7) Mancozeb has been combined with various SDHI fungicides for disease control. There is a need in the art to improve on the disease spectrum provided by these combinations.

(8) There is therefore a need in the art for combinations of SDHIs with a specific fungicide that helps improve spectrum. With crop tolerances decreasing, lower use rates being imposed and resistance being increasingly observed, there is a need for a combination of actives that allows for broader disease control spectrum that combines curative and preventive actives and has a lower dosage.

(9) Therefore, embodiments of the present invention may ameliorate one or more of the above mentioned problems.

(10) Therefore, embodiments of the present invention may provide combinations of fungicides that possess an enhanced efficacy over the individual fungicides used in isolation.

(11) Another object of the present invention is to provide a fungicidal combination that causes an enhanced greening of the crops to which it is administered.

(12) Another object of the present invention is to provide a fungicidal combination that causes late senescence to the crop to which it is applied thereby resulting into an increasing yield of the crop.

(13) Yet another object of the present invention is to provide a fungicidal combination that results into reduced fungal disease incidence in the crops to which it is applied.

(14) Another object of the present invention is to provide a fungicidal combination that achieves increased yield in the crops to which it is applied.

(15) Some or all these and other objects of the invention are can be achieved by way of the invention described hereinafter.

SUMMARY OF THE INVENTION

(16) Thus, an aspect of the present invention can provide a fungicidal combination comprising at least one succinate dehydrogenase inhibitor fungicide, at least one dithiocarbamate fungicide, and at least another fungicide.

(17) Another aspect of the present invention can provide a fungicidal combination comprising at least one succinate dehydrogenase inhibitor fungicide, at least one dithiocarbamate fungicide and at least two other fungicides.

(18) Another aspect of the present invention can provide synergistic compositions comprising at least one succinate dehydrogenase inhibitor fungicide, at least one dithiocarbamate fungicide, and at least one other fungicide.

(19) Another aspect of the present invention can provide synergistic compositions comprising at least one succinate dehydrogenase inhibitor fungicide, at least one dithiocarbamate fungicide and at least two other fungicides.

Description

DETAILED DESCRIPTION

(1) The term ‘disease control’ as used herein denotes control and prevention of a disease. Controlling effects include all deviation from natural development, for example: killing, retardation, decrease of the fungal disease. The term ‘plants’ refers to all physical parts of a plant, including seeds, seedlings, saplings, roots, tubers, stems, stalks, foliage and fruits. The term “locus” of a plant as used herein is intended to embrace the place on which the plants are growing, where the plant propagation materials of the plants are sown or where the plant propagation materials of the plants will be placed into the soil. The term “plant propagation material” is understood to denote generative parts of a plant, such as seeds, vegetative material such as cuttings or tubers, roots, fruits, tubers, bulbs, rhizomes and parts of plants, germinated plants and young plants which

are to be transplanted after germination or after emergence from the soil. These young plants may be protected before transplantation by a total or partial treatment by immersion. The term "agriculturally acceptable amount of active" refers to an amount of an active that kills or inhibits the plant disease for which control is desired, in an amount not significantly toxic to the plant being treated.

(2) Succinate dehydrogenase inhibitor (SDHI) fungicides play an important role in plant protection against many phytopathogenic fungi. These molecules specifically bind to the ubiquinone-binding site (Q-site) of the mitochondrial complex II, thereby inhibiting fungal respiration. Dithiocarbamate are multi-site contact fungicides. These molecules attack multiple sites within the fungal cells.

(3) It has surprisingly been found that the addition of a dithiocarbamate fungicide to the combinations of succinate dehydrogenase inhibitors with at least another fungicide selected from an ergosterol biosynthesis inhibitors or Quinone outside inhibitors or a combination thereof, resulted in surprising and unexpected advantages. It was surprising that the addition of a dithiocarbamate fungicide to the combination of a succinate dehydrogenase inhibitor with at least another fungicide selected from an ergosterol biosynthesis inhibitors and/or Quinone outside inhibitors or a combination thereof resulted in an enhancement of the efficacy, and a surprising reduction in fungal disease incidence, seen only with the combination of succinate dehydrogenase inhibitors with at least another fungicide selected from an ergosterol biosynthesis inhibitors or Quinone outside inhibitors or a combination thereof. It has further been found that the addition of a dithiocarbamate fungicide to these combinations and application of these combinations during the flowering stage of the crop delayed the senescence in the crop to which they were applied, which led to better greening in the crop thereby increasing the level of photosynthesis occurring within the plant, thereby leading to a greater yield from the crop to which they were applied.

(4) These surprising advantages of the combinations of the invention were not observed when the dithiocarbamate fungicide was not present in the combination. Therefore, these unexpected advantages of the combination of the present invention could be attributed to the inclusion of a dithiocarbamate fungicide to the combination of a succinate dehydrogenase inhibitor with at least another fungicide selected from an ergosterol biosynthesis inhibitors and/or Quinone outside inhibitors or a combination thereof.

(5) Thus, in an aspect, the present invention provides a fungicidal combination comprising: (a) at least one dithiocarbamate fungicide; (b) at least one succinate dehydrogenase inhibitor fungicide; and (c) at least another fungicide selected from an ergosterol biosynthesis inhibitors and/or Quinone outside inhibitors.

(6) In an embodiment, the dithiocarbamate fungicide is selected from the group consisting of amobam, asomate, azithiram, carbamorph, cufraneb, cuprodam, disulfiram, ferbam, metam, nabam, tecoram, thiram, urbacide, ziram, dazomet, etem, milneb, mancozeb, maneb, metiram, polycarbamate, propineb and zineb.

(7) In an embodiment, the preferred dithiocarbamate fungicide is mancozeb.

(8) In an embodiment, the succinate dehydrogenase inhibitor is selected from pyrazole carboxamide class of succinate dehydrogenase inhibitor fungicides. However, it should be understood that the choice of succinate dehydrogenase inhibitors is not understood to be limited to these pyrazole carboxamide fungicides alone.

(9) In an embodiment, the pyrazole carboxamide class of succinate dehydrogenase inhibitor fungicide may be selected from benzovindiflupyr, bixafen, fluxapyroxad, furametpyr, isopyrazam, penflufen, penthiopyrad, and sedaxane.

(10) Benzovindiflupyr has the chemical name N-[(1RS,4SR)-9-(dichloromethylene)-1,2,3,4-tetrahydro-1,4-methanonaphthalen-5-yl]-3-(difluoromethyl)-1-methylpyrazole-4-carboxamide and has the structure:

(11) ##STR00001##

(12) Bixafen has the chemical name N-(3',4'-dichloro-5-fluorobiphenyl-2-yl)-3-(difluoromethyl)-1-

methylpyrazole-4-carboxamide and the structure:

(13) ##STR00002##

(14) Fluxapyroxad has the chemical name 3-(difluoromethyl)-1-methyl-N-(3',4',5'-trifluorobiphenyl-2-yl)pyrazole-4-carboxamide and has the structure:

(15) ##STR00003##

(16) Furametpyr has the chemical name (RS)-5-chloro-N-(1,3-dihydro-1,1,3-trimethylisobenzofuran-4-yl)-1,3-dimethylpyrazole-4-carboxamide and has the structure:

(17) ##STR00004##

(18) Isopyrazam is a mixture of 2 isomers 3-(difluoromethyl)-1-methyl-N-[(1RS,4SR,9RS)-1,2,3,4-tetrahydro-9-isopropyl-1,4-methanonaphthalen-5-yl]pyrazole-4-carboxamide and 2 isomers 3-(difluoromethyl)-1-methyl-N-[(1RS,4SR,9SR)-1,2,3,4-tetrahydro-9-isopropyl-1,4-methanonaphthalen-5-yl]pyrazole-4-carboxamide and its tautomer's have the structure:

(19) ##STR00005##

(20) Penflufen has the chemical name N-[2-(1,3-dimethylbutyl)phenyl]-5-fluoro-1,3-dimethyl-1H-pyrazole-4-carboxamide, and has the following structure:

(21) ##STR00006##

(22) Penthioapyrad has the chemical name (RS)—N-[2-(1,3-dimethylbutyl)-3-thienyl]-1-methyl-3-(trifluoromethyl)pyrazole-4-carboxamide, and has the following structure:

(23) ##STR00007##

(24) Sedaxane is a mixture of 2 cis-isomers 2'-[(1RS,2RS)-1,1'-bicycloprop-2-yl]-3-(difluoromethyl)-1-methylpyrazole-4-carboxanilide and 2 trans-isomers 2'-[(1RS,2SR)-1,1'-bicycloprop-2-yl]-3-(difluoromethyl)-1-methylpyrazole-4-carboxanilide, and its tautomer's have the structure:

(25) ##STR00008##

(26) In an embodiment, the succinate dehydrogenase inhibitor fungicide may be selected from the group consisting of benodanil, flutolanil, mepronil, isofetamid, fluopyram, fenfuram, carboxin, oxycarboxin, thifluzamide, boscalid and IR9792.

(27) In an embodiment, the third and/or fourth fungicides in the combinations of the present invention may be selected from ergosterol biosynthesis inhibitors, and/or Quinone outside (Qo) inhibitors or mixtures thereof.

(28) Thus in an embodiment, the ergosterol biosynthesis inhibitors may be selected from the group consisting of azaconazole, bitertanol, bromuconazole, cyproconazole, difenoconazole, diniconazole, epoxiconazole, etaconazole, fenbuconazole, fluquinconazole, flusilazole, flutriafol, hexaconazole, imibenconazole, Ipconazole, metconazole, myclobutanil, penconazole, Propiconazole, simeconazole, tebuconazole, tetraconazole, triadimefon, triadimenol, triticonazole, prothioconazole, imazalil, oxpoconazole, pefurazoate, prochloraz, triflumizole, fenarimol, nuarimol, pyrifenoxy, pyrisoxazole, triforine and mixtures thereof.

(29) In another embodiment, the ergosterol biosynthesis inhibitors may be selected from prothioconazole, tebuconazole, hexaconazole, cyroconazole or epoxiconazole.

(30) In an embodiment, the third fungicide may be a Quinone outside (Qo) inhibitor fungicide selected from azoxystrobin, coumoxystrobin, enoxastrobin, flufenoxystrobin, picoxystrobin, pyraoxystrobin, mandestrobin, pyraclostrobin, pyrametostrobin, triclopyricarb, kresoxim-methyl, dimoxystrobin, fenaminostrobin, metominostrobin, trifloxystrobin, famoxadone, fluoxastrobin, fenamidone, pyribencarb and mixtures thereof.

(31) In an embodiment, the Quinone outside (Qo) inhibitor fungicide may be selected from azoxystrobin, picoxystrobin, kresoxim-methyl, pyraclostrobin and trifloxystrobin.

(32) In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is isopyrazam.

(33) In an embodiment, the combinations of the present invention include the following preferred combinations:

(34) TABLE-US-00001 S No. I II III IV 1 Mancozeb Isopyrazam Cyproconazole — 2 Mancozeb Isopyrazam Difenconazole — 3 Mancozeb Isopyrazam Epoxiconazole — 4 Mancozeb Isopyrazam Hexaconazole — 5 Mancozeb Isopyrazam Tebuconazole — 6 Mancozeb Isopyrazam Tetraconazole — 7 Mancozeb Isopyrazam Prothioconazole — 8 Mancozeb Isopyrazam — Azoxystrobin 9 Mancozeb Isopyrazam — Picoxystrobin 10 Mancozeb Isopyrazam — Pyraclostrobin 11 Mancozeb Isopyrazam — Kresoxim- methyl 12 Mancozeb Isopyrazam — Trifloxystrobin 13 Mancozeb Isopyrazam Cyproconazole Azoxystrobin 14 Mancozeb Isopyrazam Cyproconazole Picoxystrobin 15 Mancozeb Isopyrazam Cyproconazole Pyraclostrobin 16 Mancozeb Isopyrazam Cyproconazole Kresoxim- methyl 17 Mancozeb Isopyrazam Cyproconazole Trifloxystrobin 18 Mancozeb Isopyrazam Difenconazole Azoxystrobin 19 Mancozeb Isopyrazam Difenconazole Picoxystrobin 20 Mancozeb Isopyrazam Difenconazole Pyraclostrobin 21 Mancozeb Isopyrazam Difenconazole Kresoxim- methyl 22 Mancozeb Isopyrazam Difenconazole Trifloxystrobin 23 Mancozeb Isopyrazam Epoxiconazole Azoxystrobin 24 Mancozeb Isopyrazam Epoxiconazole Picoxystrobin 25 Mancozeb Isopyrazam Epoxiconazole Pyraclostrobin 26 Mancozeb Isopyrazam Epoxiconazole Kresoxim- methyl 27 Mancozeb Isopyrazam Epoxiconazole Trifloxystrobin 28 Mancozeb Isopyrazam Hexaconazole Azoxystrobin 29 Mancozeb Isopyrazam Hexaconazole Picoxystrobin 30 Mancozeb Isopyrazam Hexaconazole Pyraclostrobin 31 Mancozeb Isopyrazam Hexaconazole Kresoxim- methyl 32 Mancozeb Isopyrazam Hexaconazole Trifloxystrobin 33 Mancozeb Isopyrazam Tebuconazole Azoxystrobin 34 Mancozeb Isopyrazam Tebuconazole Picoxystrobin 35 Mancozeb Isopyrazam Tebuconazole Pyraclostrobin 36 Mancozeb Isopyrazam Tebuconazole Kresoxim- methyl 37 Mancozeb Isopyrazam Tebuconazole Trifloxystrobin 38 Mancozeb Isopyrazam Tetraconazole Azoxystrobin 39 Mancozeb Isopyrazam Tetraconazole Picoxystrobin 40 Mancozeb Isopyrazam Tetraconazole Pyraclostrobin 41 Mancozeb Isopyrazam Tetraconazole Kresoxim- methyl 42 Mancozeb Isopyrazam Tetraconazole Trifloxystrobin 43 Mancozeb Isopyrazam Prothioconazole Azoxystrobin 44 Mancozeb Isopyrazam Prothioconazole Picoxystrobin 45 Mancozeb Isopyrazam Prothioconazole Pyraclostrobin 46 Mancozeb Isopyrazam Prothioconazole Kresoxim- methyl 47 Mancozeb Isopyrazam Prothioconazole Trifloxystrobin

(35) In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is Benzovindiflupyr.

(36) In an embodiment, the combinations of the present invention include the following preferred combinations:

(37) TABLE-US-00002 S No. I II III IV 48 Mancozeb Benzovindiflupyr Cyproconazole — 49 Mancozeb Benzovindiflupyr Difenconazole — 50 Mancozeb Benzovindiflupyr Epoxiconazole — 51 Mancozeb Benzovindiflupyr Hexaconazole — 52 Mancozeb Benzovindiflupyr Tebuconazole — 53 Mancozeb Benzovindiflupyr Tetraconazole — 54 Mancozeb Benzovindiflupyr Prothioconazole — 55 Mancozeb Benzovindiflupyr — Azoxystrobin 56 Mancozeb Benzovindiflupyr — Picoxystrobin 57 Mancozeb Benzovindiflupyr — Pyraclostrobin 58 Mancozeb Benzovindiflupyr — Kresoxim- methyl 59 Mancozeb Benzovindiflupyr — Trifloxystrobin 60 Mancozeb Benzovindiflupyr Cyproconazole Azoxystrobin 61 Mancozeb Benzovindiflupyr Cyproconazole Picoxystrobin 62 Mancozeb Benzovindiflupyr Cyproconazole Pyraclostrobin 63 Mancozeb Benzovindiflupyr Cyproconazole Kresoxim- methyl 64 Mancozeb Benzovindiflupyr Cyproconazole Trifloxystrobin 65 Mancozeb Benzovindiflupyr Difenconazole Azoxystrobin 66 Mancozeb Benzovindiflupyr Difenconazole Picoxystrobin 67 Mancozeb Benzovindiflupyr Difenconazole Pyraclostrobin 68 Mancozeb Benzovindiflupyr Difenconazole Kresoxim- methyl 69 Mancozeb Benzovindiflupyr Difenconazole Trifloxystrobin 70 Mancozeb Benzovindiflupyr Epoxiconazole Azoxystrobin 71 Mancozeb Benzovindiflupyr Epoxiconazole Picoxystrobin 72 Mancozeb Benzovindiflupyr Epoxiconazole Pyraclostrobin 73 Mancozeb Benzovindiflupyr Epoxiconazole Kresoxim- methyl 74 Mancozeb Benzovindiflupyr Epoxiconazole Trifloxystrobin 75 Mancozeb Benzovindiflupyr Hexaconazole Azoxystrobin 76 Mancozeb Benzovindiflupyr

Hexaconazole Picoxystrobin 77 Mancozeb Benzovindiflupyr Hexaconazole Pyraclostrobin 78 Mancozeb Benzovindiflupyr Hexaconazole Kresoxim- methyl 79 Mancozeb Benzovindiflupyr Hexaconazole Trifloxystrobin 80 Mancozeb Benzovindiflupyr Tebuconazole Azoxystrobin 81 Mancozeb Benzovindiflupyr Tebuconazole Picoxystrobin 82 Mancozeb Benzovindiflupyr Tebuconazole Pyraclostrobin 83 Mancozeb Benzovindiflupyr Tebuconazole Kresoxim- methyl 84 Mancozeb Benzovindiflupyr Tebuconazole Trifloxystrobin 85 Mancozeb Benzovindiflupyr Tetraconazole Azoxystrobin 86 Mancozeb Benzovindiflupyr Tetraconazole Picoxystrobin 87 Mancozeb Benzovindiflupyr Tetraconazole Pyraclostrobin 88 Mancozeb Benzovindiflupyr Tetraconazole Kresoxim- methyl 89 Mancozeb Benzovindiflupyr Tetraconazole Trifloxystrobin 90 Mancozeb Benzovindiflupyr Prothioconazole Azoxystrobin 91 Mancozeb Benzovindiflupyr Prothioconazole Picoxystrobin 92 Mancozeb Benzovindiflupyr Prothioconazole Pyraclostrobin 93 Mancozeb Benzovindiflupyr Prothioconazole Kresoxim- methyl 94 Mancozeb Benzovindiflupyr Prothioconazole Trifloxystrobin

(38) In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is Penthiopyrad.

(39) In an embodiment, the combinations of the present invention include the following preferred combinations:

(40) TABLE-US-00003 S No. I II III IV 95 Mancozeb Penthiopyrad Cyproconazole — 96 Mancozeb Penthiopyrad Difenconazole — 97 Mancozeb Penthiopyrad Epoxiconazole — 98 Mancozeb Penthiopyrad Hexaconazole — 99 Mancozeb Penthiopyrad Tebuconazole — 100 Mancozeb Penthiopyrad Tetraconazole — 101 Mancozeb Penthiopyrad Prothioconazole — 102 Mancozeb Penthiopyrad — Azoxystrobin 103 Mancozeb Penthiopyrad — Picoxystrobin 104 Mancozeb Penthiopyrad — Pyraclostrobin 105 Mancozeb Penthiopyrad — Kresoxim- methyl 106 Mancozeb Penthiopyrad — Trifloxystrobin 107 Mancozeb Penthiopyrad Cyproconazole Azoxystrobin 108 Mancozeb Penthiopyrad Cyproconazole Picoxystrobin 109 Mancozeb Penthiopyrad Cyproconazole Pyraclostrobin 110 Mancozeb Penthiopyrad Cyproconazole Kresoxim- methyl 111 Mancozeb Penthiopyrad Cyproconazole Trifloxystrobin 112 Mancozeb Penthiopyrad Difenconazole Azoxystrobin 113 Mancozeb Penthiopyrad Difenconazole Picoxystrobin 114 Mancozeb Penthiopyrad Difenconazole Pyraclostrobin 115 Mancozeb Penthiopyrad Difenconazole Kresoxim- methyl 116 Mancozeb Penthiopyrad Difenconazole Trifloxystrobin 117 Mancozeb Penthiopyrad Epoxiconazole Azoxystrobin 118 Mancozeb Penthiopyrad Epoxiconazole Picoxystrobin 119 Mancozeb Penthiopyrad Epoxiconazole Pyraclostrobin 120 Mancozeb Penthiopyrad Epoxiconazole Kresoxim- methyl 121 Mancozeb Penthiopyrad Epoxiconazole Trifloxystrobin 122 Mancozeb Penthiopyrad Hexaconazole Azoxystrobin 123 Mancozeb Penthiopyrad Hexaconazole Picoxystrobin 124 Mancozeb Penthiopyrad Hexaconazole Pyraclostrobin 125 Mancozeb Penthiopyrad Hexaconazole Kresoxim- methyl 126 Mancozeb Penthiopyrad Hexaconazole Trifloxystrobin 127 Mancozeb Penthiopyrad Tebuconazole Azoxystrobin 128 Mancozeb Penthiopyrad Tebuconazole Picoxystrobin 129 Mancozeb Penthiopyrad Tebuconazole Pyraclostrobin 130 Mancozeb Penthiopyrad Tebuconazole Kresoxim- methyl 131 Mancozeb Penthiopyrad Tebuconazole Trifloxystrobin 132 Mancozeb Penthiopyrad Tetraconazole Azoxystrobin 133 Mancozeb Penthiopyrad Tetraconazole Picoxystrobin 134 Mancozeb Penthiopyrad Tetraconazole Pyraclostrobin 135 Mancozeb Penthiopyrad Tetraconazole Kresoxim- methyl 136 Mancozeb Penthiopyrad Tetraconazole Trifloxystrobin 137 Mancozeb Penthiopyrad Prothioconazole Azoxystrobin 138 Mancozeb Penthiopyrad Prothioconazole Picoxystrobin 139 Mancozeb Penthiopyrad Prothioconazole Pyraclostrobin 140 Mancozeb Penthiopyrad Prothioconazole Kresoxim- methyl 141 Mancozeb Penthiopyrad Prothioconazole Trifloxystrobin

(41) In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is boscalid.

(42) In an embodiment, the combinations of the present invention include the following preferred

combinations:

(43) TABLE-US-00004 S No. I II III IV 142 Mancozeb Boscalid Cyproconazole — 143 Mancozeb Boscalid Difenconazole — 144 Mancozeb Boscalid Epoxiconazole — 145 Mancozeb Boscalid Hexaconazole — 146 Mancozeb Boscalid Tebuconazole — 147 Mancozeb Boscalid Tetraconazole — 148 Mancozeb Boscalid Prothioconazole — 149 Mancozeb Boscalid — Azoxystrobin 150 Mancozeb Boscalid — Picoxystrobin 151 Mancozeb Boscalid — Pyraclostrobin 152 Mancozeb Boscalid — Kresoxim- methyl 153 Mancozeb Boscalid — Trifloxystrobin 154 Mancozeb Boscalid Cyproconazole Azoxystrobin 155 Mancozeb Boscalid Cyproconazole Picoxystrobin 156 Mancozeb Boscalid Cyproconazole Pyraclostrobin 157 Mancozeb Boscalid Cyproconazole Kresoxim- methyl 158 Mancozeb Boscalid Cyproconazole Trifloxystrobin 159 Mancozeb Boscalid Difenconazole Azoxystrobin 160 Mancozeb Boscalid Difenconazole Picoxystrobin 161 Mancozeb Boscalid Difenconazole Pyraclostrobin 162 Mancozeb Boscalid Difenconazole Kresoxim- methyl 163 Mancozeb Boscalid Difenconazole Trifloxystrobin 164 Mancozeb Boscalid Epoxiconazole Azoxystrobin 165 Mancozeb Boscalid Epoxiconazole Picoxystrobin 166 Mancozeb Boscalid Epoxiconazole Pyraclostrobin 167 Mancozeb Boscalid Epoxiconazole Kresoxim- methyl 168 Mancozeb Boscalid Epoxiconazole Trifloxystrobin 169 Mancozeb Boscalid Hexaconazole Azoxystrobin 170 Mancozeb Boscalid Hexaconazole Picoxystrobin 171 Mancozeb Boscalid Hexaconazole Pyraclostrobin 172 Mancozeb Boscalid Hexaconazole Kresoxim- methyl 173 Mancozeb Boscalid Hexaconazole Trifloxystrobin 174 Mancozeb Boscalid Tebuconazole Azoxystrobin 175 Mancozeb Boscalid Tebuconazole Picoxystrobin 176 Mancozeb Boscalid Tebuconazole Pyraclostrobin 177 Mancozeb Boscalid Tebuconazole Kresoxim- methyl 178 Mancozeb Boscalid Tebuconazole Trifloxystrobin 179 Mancozeb Boscalid Tetraconazole Azoxystrobin 180 Mancozeb Boscalid Tetraconazole Picoxystrobin 181 Mancozeb Boscalid Tetraconazole Pyraclostrobin 182 Mancozeb Boscalid Tetraconazole Kresoxim- methyl 183 Mancozeb Boscalid Tetraconazole Trifloxystrobin 184 Mancozeb Boscalid Prothioconazole Azoxystrobin 185 Mancozeb Boscalid Prothioconazole Picoxystrobin 186 Mancozeb Boscalid Prothioconazole Pyraclostrobin 187 Mancozeb Boscalid Prothioconazole Kresoxim- methyl 188 Mancozeb Boscalid Prothioconazole Trifloxystrobin

(44) In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is IR9792. IR9792 is a SDHI fungicide, developed by Isagro, with the proposed common name fluindapyr.

(45) In an embodiment, the combinations of the present invention include the following preferred combinations:

(46) TABLE-US-00005 S No. I II III IV 189 Mancozeb IR9792 Cyproconazole — 190 Mancozeb IR9792 Difenconazole — 191 Mancozeb IR9792 Epoxiconazole — 192 Mancozeb IR9792 Hexaconazole — 193 Mancozeb IR9792 Tebuconazole — 194 Mancozeb IR9792 Tetraconazole — 195 Mancozeb IR9792 Prothioconazole — 196 Mancozeb IR9792 — Azoxystrobin 197 Mancozeb IR9792 — Picoxystrobin 198 Mancozeb IR9792 — Pyraclostrobin 199 Mancozeb IR9792 — Kresoxim- methyl 200 Mancozeb IR9792 — Trifloxystrobin 201 Mancozeb IR9792 Cyproconazole Azoxystrobin 202 Mancozeb IR9792 Cyproconazole Picoxystrobin 203 Mancozeb IR9792 Cyproconazole Pyraclostrobin 204 Mancozeb IR9792 Cyproconazole Kresoxim- methyl 205 Mancozeb IR9792 Cyproconazole Trifloxystrobin 206 Mancozeb IR9792 Difenconazole Azoxystrobin 207 Mancozeb IR9792 Difenconazole Picoxystrobin 208 Mancozeb IR9792 Difenconazole Pyraclostrobin 209 Mancozeb IR9792 Difenconazole Kresoxim- methyl 210 Mancozeb IR9792 Difenconazole Trifloxystrobin 211 Mancozeb IR9792 Epoxiconazole Azoxystrobin 212 Mancozeb IR9792 Epoxiconazole Picoxystrobin 213 Mancozeb IR9792 Epoxiconazole Pyraclostrobin 214 Mancozeb IR9792 Epoxiconazole Kresoxim- methyl 215 Mancozeb IR9792 Epoxiconazole Trifloxystrobin 216 Mancozeb IR9792 Hexaconazole Azoxystrobin 217 Mancozeb IR9792 Hexaconazole Picoxystrobin 218 Mancozeb IR9792 Hexaconazole Pyraclostrobin 219 Mancozeb IR9792 Hexaconazole Kresoxim- methyl 220

Mancozeb IR9792 Hexaconazole Trifloxystrobin 221 Mancozeb IR9792 Tebuconazole
Azoxystrobin 222 Mancozeb IR9792 Tebuconazole Picoxystrobin 223 Mancozeb IR9792
Tebuconazole Pyraclostrobin 224 Mancozeb IR9792 Tebuconazole Kresoxim- methyl 225
Mancozeb IR9792 Tebuconazole Trifloxystrobin 226 Mancozeb IR9792 Tetraconazole
Azoxystrobin 227 Mancozeb IR9792 Tetraconazole Picoxystrobin 228 Mancozeb IR9792
Tetraconazole Pyraclostrobin 229 Mancozeb IR9792 Tetraconazole Kresoxim- methyl 230
Mancozeb IR9792 Tetraconazole Trifloxystrobin 231 Mancozeb IR9792 Prothioconazole
Azoxystrobin 232 Mancozeb IR9792 Prothioconazole Picoxystrobin 233 Mancozeb IR9792
Prothioconazole Pyraclostrobin 234 Mancozeb IR9792 Prothioconazole Kresoxim- methyl 235
Mancozeb IR9792 Prothioconazole Trifloxystrobin

(47) In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is Bixafen.

(48) In an embodiment, the combinations of the present invention include the following preferred combinations:

(49) TABLE-US-00006 S No. I II III IV 236 Mancozeb Bixafen Cyproconazole — 237 Mancozeb Bixafen Difenconazole — 238 Mancozeb Bixafen Epoxiconazole — 239 Mancozeb Bixafen Hexaconazole — 240 Mancozeb Bixafen Tebuconazole — 241 Mancozeb Bixafen Tetraconazole — 242 Mancozeb Bixafen Prothioconazole — 243 Mancozeb Bixafen — Azoxystrobin 244 Mancozeb Bixafen — Picoxystrobin 245 Mancozeb Bixafen — Pyraclostrobin 246 Mancozeb Bixafen — Kresoxim- methyl 247 Mancozeb Bixafen — Trifloxystrobin 248 Mancozeb Bixafen Cyproconazole Azoxystrobin 249 Mancozeb Bixafen Cyproconazole Picoxystrobin 250 Mancozeb Bixafen Cyproconazole Pyraclostrobin 251 Mancozeb Bixafen Cyproconazole Kresoxim- methyl 252 Mancozeb Bixafen Cyproconazole Trifloxystrobin 253 Mancozeb Bixafen Difenconazole Azoxystrobin 254 Mancozeb Bixafen Difenconazole Picoxystrobin 255 Mancozeb Bixafen Difenconazole Pyraclostrobin 256 Mancozeb Bixafen Difenconazole Kresoxim- methyl 257 Mancozeb Bixafen Difenconazole Trifloxystrobin 258 Mancozeb Bixafen Epoxiconazole Azoxystrobin 259 Mancozeb Bixafen Epoxiconazole Picoxystrobin 260 Mancozeb Bixafen Epoxiconazole Pyraclostrobin 261 Mancozeb Bixafen Epoxiconazole Kresoxim- methyl 262 Mancozeb Bixafen Epoxiconazole Trifloxystrobin 263 Mancozeb Bixafen Hexaconazole Azoxystrobin 264 Mancozeb Bixafen Hexaconazole Picoxystrobin 265 Mancozeb Bixafen Hexaconazole Pyraclostrobin 266 Mancozeb Bixafen Hexaconazole Kresoxim- methyl 267 Mancozeb Bixafen Hexaconazole Trifloxystrobin 268 Mancozeb Bixafen Tebuconazole Azoxystrobin 269 Mancozeb Bixafen Tebuconazole Picoxystrobin 270 Mancozeb Bixafen Tebuconazole Pyraclostrobin 271 Mancozeb Bixafen Tebuconazole Kresoxim- methyl 272 Mancozeb Bixafen Tebuconazole Trifloxystrobin 273 Mancozeb Bixafen Tetraconazole Azoxystrobin 274 Mancozeb Bixafen Tetraconazole Picoxystrobin 275 Mancozeb Bixafen Tetraconazole Pyraclostrobin 276 Mancozeb Bixafen Tetraconazole Kresoxim- methyl 277 Mancozeb Bixafen Tetraconazole Trifloxystrobin 278 Mancozeb Bixafen Prothioconazole Azoxystrobin 279 Mancozeb Bixafen Prothioconazole Picoxystrobin 280 Mancozeb Bixafen Prothioconazole Pyraclostrobin 281 Mancozeb Bixafen Prothioconazole Kresoxim- methyl 282 Mancozeb Bixafen Prothioconazole Trifloxystrobin

(50) In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is Fluxapyroxad.

(51) In an embodiment, the combinations of the present invention include the following preferred combinations:

(52) TABLE-US-00007 S No. I II III IV 283 Mancozeb Fluxapyroxad Cyproconazole — 284 Mancozeb Fluxapyroxad Difenconazole — 285 Mancozeb Fluxapyroxad Epoxiconazole — 286 Mancozeb Fluxapyroxad Hexaconazole — 287 Mancozeb Fluxapyroxad Tebuconazole — 288 Mancozeb Fluxapyroxad Tetraconazole — 289 Mancozeb Fluxapyroxad Prothioconazole — 290 Mancozeb Fluxapyroxad — Azoxystrobin 291 Mancozeb Fluxapyroxad — Picoxystrobin 292

Mancozeb Fluxapyroxad — Pyraclostrobin 293 Mancozeb Fluxapyroxad — Kresoxim- methyl 294
 Mancozeb Fluxapyroxad — Trifloxystrobin 295 Mancozeb Fluxapyroxad Cyproconazole
 Azoxystrobin 296 Mancozeb Fluxapyroxad Cyproconazole Picoxystrobin 297 Mancozeb
 Fluxapyroxad Cyproconazole Pyraclostrobin 298 Mancozeb Fluxapyroxad Cyproconazole
 Kresoxim- methyl 299 Mancozeb Fluxapyroxad Cyproconazole Trifloxystrobin 300 Mancozeb
 Fluxapyroxad Difenconazole Azoxystrobin 301 Mancozeb Fluxapyroxad Difenconazole
 Picoxystrobin 302 Mancozeb Fluxapyroxad Difenconazole Pyraclostrobin 303 Mancozeb
 Fluxapyroxad Difenconazole Kresoxim- methyl 304 Mancozeb Fluxapyroxad Difenconazole
 Trifloxystrobin 305 Mancozeb Fluxapyroxad Epoxiconazole Azoxystrobin 306 Mancozeb
 Fluxapyroxad Epoxiconazole Picoxystrobin 307 Mancozeb Fluxapyroxad Epoxiconazole
 Pyraclostrobin 308 Mancozeb Fluxapyroxad Epoxiconazole Kresoxim- methyl 309 Mancozeb
 Fluxapyroxad Epoxiconazole Trifloxystrobin 310 Mancozeb Fluxapyroxad Hexaconazole
 Azoxystrobin 311 Mancozeb Fluxapyroxad Hexaconazole Picoxystrobin 312 Mancozeb
 Fluxapyroxad Hexaconazole Pyraclostrobin 313 Mancozeb Fluxapyroxad Hexaconazole
 Kresoxim- methyl 314 Mancozeb Fluxapyroxad Hexaconazole Trifloxystrobin 315 Mancozeb
 Fluxapyroxad Tebuconazole Azoxystrobin 316 Mancozeb Fluxapyroxad Tebuconazole
 Picoxystrobin 317 Mancozeb Fluxapyroxad Tebuconazole Pyraclostrobin 318 Mancozeb
 Fluxapyroxad Tebuconazole Kresoxim- methyl 319 Mancozeb Fluxapyroxad Tebuconazole
 Trifloxystrobin 320 Mancozeb Fluxapyroxad Tetraconazole Azoxystrobin 321 Mancozeb
 Fluxapyroxad Tetraconazole Picoxystrobin 322 Mancozeb Fluxapyroxad Tetraconazole
 Pyraclostrobin 323 Mancozeb Fluxapyroxad Tetraconazole Kresoxim- methyl 324 Mancozeb
 Fluxapyroxad Tetraconazole Trifloxystrobin 325 Mancozeb Fluxapyroxad Prothioconazole
 Azoxystrobin 326 Mancozeb Fluxapyroxad Prothioconazole Picoxystrobin 327 Mancozeb
 Fluxapyroxad Prothioconazole Pyraclostrobin 328 Mancozeb Fluxapyroxad Prothioconazole
 Kresoxim- methyl 329 Mancozeb Fluxapyroxad Prothioconazole Trifloxystrobin

(53) In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is Furametpyr.

(54) In an embodiment, the combinations of the present invention include the following preferred combinations:

(55) TABLE-US-00008 S No. I II III IV 330 Mancozeb Furametpyr Cyproconazole — 331
 Mancozeb Furametpyr Difenconazole — 332 Mancozeb Furametpyr Epoxiconazole — 333
 Mancozeb Furametpyr Hexaconazole — 334 Mancozeb Furametpyr Tebuconazole — 335
 Mancozeb Furametpyr Tetraconazole — 336 Mancozeb Furametpyr Prothioconazole — 337
 Mancozeb Furametpyr — Azoxystrobin 338 Mancozeb Furametpyr — Picoxystrobin 339
 Mancozeb Furametpyr — Pyraclostrobin 340 Mancozeb Furametpyr — Kresoxim- methyl 341
 Mancozeb Furametpyr — Trifloxystrobin 342 Mancozeb Furametpyr Cyproconazole Azoxystrobin
 343 Mancozeb Furametpyr Cyproconazole Picoxystrobin 344 Mancozeb Furametpyr
 Cyproconazole Pyraclostrobin 345 Mancozeb Furametpyr Cyproconazole Kresoxim- methyl 346
 Mancozeb Furametpyr Cyproconazole Trifloxystrobin 347 Mancozeb Furametpyr Difenconazole
 Azoxystrobin 348 Mancozeb Furametpyr Difenconazole Picoxystrobin 349 Mancozeb
 Furametpyr Difenconazole Pyraclostrobin 350 Mancozeb Furametpyr Difenconazole Kresoxim-
 methyl 351 Mancozeb Furametpyr Difenconazole Trifloxystrobin 352 Mancozeb Furametpyr
 Epoxiconazole Azoxystrobin 353 Mancozeb Furametpyr Epoxiconazole Picoxystrobin 354
 Mancozeb Furametpyr Epoxiconazole Pyraclostrobin 355 Mancozeb Furametpyr Epoxiconazole
 Kresoxim- methyl 356 Mancozeb Furametpyr Epoxiconazole Trifloxystrobin 357 Mancozeb
 Furametpyr Hexaconazole Azoxystrobin 358 Mancozeb Furametpyr Hexaconazole Picoxystrobin
 359 Mancozeb Furametpyr Hexaconazole Pyraclostrobin 360 Mancozeb Furametpyr Hexaconazole
 Kresoxim- methyl 361 Mancozeb Furametpyr Hexaconazole Trifloxystrobin 362 Mancozeb
 Furametpyr Tebuconazole Azoxystrobin 363 Mancozeb Furametpyr Tebuconazole Picoxystrobin
 364 Mancozeb Furametpyr Tebuconazole Pyraclostrobin 365 Mancozeb Furametpyr Tebuconazole

Kresoxim- methyl 366 Mancozeb Furametpyr Tebuconazole Trifloxystrobin 367 Mancozeb Furametpyr Tetraconazole Azoxystrobin 368 Mancozeb Furametpyr Tetraconazole Picoxystrobin 369 Mancozeb Furametpyr Tetraconazole Pyraclostrobin 370 Mancozeb Furametpyr Tetraconazole Kresoxim- methyl 371 Mancozeb Furametpyr Tetraconazole Trifloxystrobin 372 Mancozeb Furametpyr Prothioconazole Azoxystrobin 273 Mancozeb Furametpyr Prothioconazole Picoxystrobin 374 Mancozeb Furametpyr Prothioconazole Pyraclostrobin 375 Mancozeb Furametpyr Prothioconazole Kresoxim- methyl 376 Mancozeb Furametpyr Prothioconazole Trifloxystrobin

(56) In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is Penflufen.

(57) In an embodiment, the combinations of the present invention include the following preferred combinations:

(58) TABLE-US-00009 S No. I II III IV 377 Mancozeb Penflufen Cyproconazole — 378 Mancozeb Penflufen Difenconazole — 379 Mancozeb Penflufen Epoxiconazole — 380 Mancozeb Penflufen Hexaconazole — 381 Mancozeb Penflufen Tebuconazole — 382 Mancozeb Penflufen Tetraconazole — 383 Mancozeb Penflufen Prothioconazole — 384 Mancozeb Penflufen — Azoxystrobin 385 Mancozeb Penflufen — Picoxystrobin 386 Mancozeb Penflufen — Pyraclostrobin 387 Mancozeb Penflufen — Kresoxim- methyl 388 Mancozeb Penflufen — Trifloxystrobin 389 Mancozeb Penflufen Cyproconazole Azoxystrobin 390 Mancozeb Penflufen Cyproconazole Picoxystrobin 391 Mancozeb Penflufen Cyproconazole Pyraclostrobin 392 Mancozeb Penflufen Cyproconazole Kresoxim- methyl 393 Mancozeb Penflufen Cyproconazole Trifloxystrobin 394 Mancozeb Penflufen Difenconazole Azoxystrobin 395 Mancozeb Penflufen Difenconazole Picoxystrobin 396 Mancozeb Penflufen Difenconazole Pyraclostrobin 397 Mancozeb Penflufen Difenconazole Kresoxim- methyl 398 Mancozeb Penflufen Difenconazole Trifloxystrobin 399 Mancozeb Penflufen Epoxiconazole Azoxystrobin 400 Mancozeb Penflufen Epoxiconazole Picoxystrobin 401 Mancozeb Penflufen Epoxiconazole Pyraclostrobin 402 Mancozeb Penflufen Epoxiconazole Kresoxim- methyl 403 Mancozeb Penflufen Epoxiconazole Trifloxystrobin 404 Mancozeb Penflufen Hexaconazole Azoxystrobin 405 Mancozeb Penflufen Hexaconazole Picoxystrobin 406 Mancozeb Penflufen Hexaconazole Pyraclostrobin 407 Mancozeb Penflufen Hexaconazole Kresoxim- methyl 408 Mancozeb Penflufen Hexaconazole Trifloxystrobin 409 Mancozeb Penflufen Tebuconazole Azoxystrobin 410 Mancozeb Penflufen Tebuconazole Picoxystrobin 411 Mancozeb Penflufen Tebuconazole Pyraclostrobin 412 Mancozeb Penflufen Tebuconazole Kresoxim- methyl 413 Mancozeb Penflufen Tebuconazole Trifloxystrobin 414 Mancozeb Penflufen Tetraconazole Azoxystrobin 415 Mancozeb Penflufen Tetraconazole Picoxystrobin 416 Mancozeb Penflufen Tetraconazole Pyraclostrobin 417 Mancozeb Penflufen Tetraconazole Kresoxim- methyl 418 Mancozeb Penflufen Tetraconazole Trifloxystrobin 419 Mancozeb Penflufen Prothioconazole Azoxystrobin 420 Mancozeb Penflufen Prothioconazole Picoxystrobin 421 Mancozeb Penflufen Prothioconazole Pyraclostrobin 422 Mancozeb Penflufen Prothioconazole Kresoxim- methyl 423 Mancozeb Penflufen Prothioconazole Trifloxystrobin

(59) In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is Sedaxane.

(60) In an embodiment, the combinations of the present invention include the following preferred combinations:

(61) TABLE-US-00010 S No. I II III IV 424 Mancozeb Sedaxane Cyproconazole — 425 Mancozeb Sedaxane Difenconazole — 426 Mancozeb Sedaxane Epoxiconazole — 427 Mancozeb Sedaxane Hexaconazole — 428 Mancozeb Sedaxane Tebuconazole — 429 Mancozeb Sedaxane Tetraconazole — 430 Mancozeb Sedaxane Prothioconazole — 431 Mancozeb Sedaxane — Azoxystrobin 432 Mancozeb Sedaxane — Picoxystrobin 433 Mancozeb Sedaxane — Pyraclostrobin 434 Mancozeb Sedaxane — Kresoxim- methyl 435 Mancozeb Sedaxane — Trifloxystrobin 436 Mancozeb Sedaxane Cyproconazole Azoxystrobin 437 Mancozeb Sedaxane

Cyproconazole Picoxystrobin 438 Mancozeb Sedaxane Cyproconazole Pyraclostrobin 439
 Mancozeb Sedaxane Cyproconazole Kresoxim- methyl 440 Mancozeb Sedaxane Cyproconazole
 Trifloxystrobin 441 Mancozeb Sedaxane Difenconazole Azoxystrobin 442 Mancozeb Sedaxane
 Difenconazole Picoxystrobin 443 Mancozeb Sedaxane Difenconazole Pyraclostrobin 444
 Mancozeb Sedaxane Difenconazole Kresoxim- methyl 445 Mancozeb Sedaxane Difenconazole
 Trifloxystrobin 446 Mancozeb Sedaxane Epoxiconazole Azoxystrobin 447 Mancozeb Sedaxane
 Epoxiconazole Picoxystrobin 448 Mancozeb Sedaxane Epoxiconazole Pyraclostrobin 449
 Mancozeb Sedaxane Epoxiconazole Kresoxim- methyl 450 Mancozeb Sedaxane Epoxiconazole
 Trifloxystrobin 451 Mancozeb Sedaxane Hexaconazole Azoxystrobin 452 Mancozeb Sedaxane
 Hexaconazole Picoxystrobin 453 Mancozeb Sedaxane Hexaconazole Pyraclostrobin 454 Mancozeb
 Sedaxane Hexaconazole Kresoxim- methyl 455 Mancozeb Sedaxane Hexaconazole Trifloxystrobin
 456 Mancozeb Sedaxane Tebuconazole Azoxystrobin 457 Mancozeb Sedaxane Tebuconazole
 Picoxystrobin 458 Mancozeb Sedaxane Tebuconazole Pyraclostrobin 459 Mancozeb Sedaxane
 Tebuconazole Kresoxim- methyl 460 Mancozeb Sedaxane Tebuconazole Trifloxystrobin 461
 Mancozeb Sedaxane Tetraconazole Azoxystrobin 462 Mancozeb Sedaxane Tetraconazole
 Picoxystrobin 463 Mancozeb Sedaxane Tetraconazole Pyraclostrobin 464 Mancozeb Sedaxane
 Tetraconazole Kresoxim- methyl 465 Mancozeb Sedaxane Tetraconazole Trifloxystrobin 466
 Mancozeb Sedaxane Prothioconazole Azoxystrobin 467 Mancozeb Sedaxane Prothioconazole
 Picoxystrobin 468 Mancozeb Sedaxane Prothioconazole Pyraclostrobin 469 Mancozeb Sedaxane
 Prothioconazole Kresoxim- methyl 470 Mancozeb Sedaxane Prothioconazole Trifloxystrobin

(62) The combinations of the present invention may be formulated in the form of a composition.

(63) In an embodiment, the present invention may provide a composition comprising: (a) at least one succinate dehydrogenase inhibitor fungicide; (b) at least one dithiocarbamate fungicide; (c) at least one quinone outside inhibitor; and (d) at least one agrochemically acceptable excipient.

(64) In an embodiment, the present invention may provide a composition comprising: (a) at least one succinate dehydrogenase inhibitor fungicide; (b) at least one dithiocarbamate fungicide; (c) at least one ergostrol biosynthesis inhibitor; and (d) at least one agrochemically acceptable excipient.

(65) In an embodiment, the present invention may provide a composition comprising: (a) at least one succinate dehydrogenase inhibitor fungicide; (b) at least one dithiocarbamate fungicide; (c) at least one a quinone outside inhibitor; (d) at least one ergostrol biosynthesis inhibitor; and (e) at least one agrochemically acceptable excipient.

(66) The amount of a composition according to the invention to be applied, will depend on various factors, such as the subject of the treatment, such as, for example plants, soil or seeds; the type of treatment, such as, for example spraying, dusting or seed dressing; the purpose of the treatment, such as, for example prophylactic or therapeutic disease control; in case of disease control the type of fungi to be controlled or the application time. This amount of the combinations of the present invention to be applied can be readily deduced by a skilled agronomist.

(67) Thus in an embodiment, the present invention may provide compositions comprising: (a) at least one pyrazaole carboxamide selected from benzovindiflupyr, bixafen, fluxapyroxad, furametpyr, isopyrazam, penflufen, penthiopyrad, and sedaxane; (b) at least one quinone outside inhibitor, and/or at least one ergostrol biosynthesis inhibitor; and (c) at last one dithiocarbamate fungicide;

said fungicides being combined in agrochemically acceptable amounts.

(68) In an embodiment, the total amount of succinate dehydrogenase inhibitor in the composition may typically be in the range of 0.1 to 99% by weight, preferably 0.2 to 90% by weight. The total amount of dithiocarbamate fungicide in the composition may be in the range of 0.1 to 99% by weight. The total amount of ergostrol biosynthesis inhibitor in the composition may be in the range of 0.1 to 99% by weight. The total amount of Quinone outside inhibitor in the composition may be in the range of 0.1 to 99% by weight.

(69) In an embodiment, the constituent fungicides of the combination of the present invention may

be admixed in ratio of (1-80):(1-80):(1-80) of the dithiocarbamate fungicide, succinate dehydrogenase inhibitor fungicide and the third fungicide respectively.

(70) In an embodiment, the constituents of the composition of the present invention may be tank mixed and sprayed at the locus of the infection, or may be alternatively be mixed with surfactants and then sprayed.

(71) In an embodiment, the constituents of the composition of the present invention may be used for foliar application, ground or applications to plant propagation materials.

(72) In an embodiment, the compositions of the present invention may typically be produce by mixing the actives in the composition with an inert carrier, and adding surfactants and other adjuvants and carriers as needed and formulated into solid, or liquid formulations, including but not limited to wettable powders, granules, dusts, Soluble (liquid) concentrates, suspension concentrates, oil in water emulsion, water in oil emulsion, emulsifiable concentrates, capsule suspensions, ZC formulations, oil dispersions or other known formulation types. The composition may also be used for treatment of a plant propagation material such as seeds etc.

(73) Examples of the solid carrier used in formulation include fine powders or granules such as minerals such as kaolin clay, attapulgitic clay, bentonite, montmorillonite, acid white clay, pyrophyllite, talc, diatomaceous earth and calcite; natural organic materials such as corn rachis powder and walnut husk powder; synthetic organic materials such as urea; salts such as calcium carbonate and ammonium sulfate; synthetic inorganic materials such as synthetic hydrated silicon oxide; and as a liquid carrier, aromatic hydrocarbons such as xylene, alkylbenzene and methylnaphthalene; alcohols such as 2-propanol, ethyleneglycol, propylene glycol, and ethylene glycol monoethyl ether; ketones such as acetone, cyclohexanone and isophorone; vegetable oil such as soybean oil and cotton seed oil; petroleum aliphatic hydrocarbons, esters, dimethylsulfoxide, acetonitrile and water.

(74) Examples of the surfactant include anionic surfactants such as alkyl sulfate ester salts, alkylaryl sulfonate salts, dialkyl sulfosuccinate salts, polyoxyethylene alkylaryl ether phosphate ester salts, lignosulfonate salts and naphthalene sulfonate formaldehyde polycondensates; and nonionic surfactants such as polyoxyethylene alkyl aryl ethers, polyoxyethylene alkylpolyoxypropylene block copolymers and sorbitan fatty acid esters and cationic surfactants such as alkyltrimethylammonium salts.

(75) Examples of the other formulation auxiliary agents include water-soluble polymers such as polyvinyl alcohol and polyvinylpyrrolidone, polysaccharides such as Arabic gum, alginic acid and the salt thereof, CMC (carboxymethyl-cellulose), Xanthan gum, inorganic materials such as aluminum magnesium silicate and alumina sol, preservatives, coloring agents and stabilization agents such as PAP (acid phosphate isopropyl) and BHT.

(76) The compositions according to the present invention is effective for the following plant diseases: Disease in rice: Blast (*Magnaporthe grisea*), Helminthosporium leaf spot (*Cochliobolus miyabeanus*), sheath blight (*Rhizoctonia solani*), and bakanae disease (*Gibberella fujikuroi*). Diseases in wheat: powdery mildew (*Erysiphe graminis*), Fusarium head blight (*Fusarium graminearum*, *F. avenacerum*, *F. culmorum*, *Microdochium nivale*), rust (*Puccinia striiformis*, *P. graminis*, *P. recondita*), pink snow mold (*Micronectriella nivale*), *Typhula* snow blight (*Typhula* sp.), loose smut (*Ustilago tritici*), bunt (*Tilletia caries*), eyespot (*Pseudocercospora herpotrichoides*), leaf blotch (*Mycosphaerella graminicola*), glume blotch (*Stagonospora nodorum*), septoria, and yellow spot (*Pyrenophora tritici-repentis*) Diseases of barley: powdery mildew (*Erysiphe graminis*), *Fusarium* head blight (*Fusarium graminearum*, *F. avenacerum*, *F. culmorum*, *Microdochium nivale*), rust (*Puccinia striiformis*, *P. graminis*, *P. hordei*), loose smut (*Ustilago nuda*), scald (*Rhynchosporium secalis*), net blotch (*Pyrenophora teres*), spot blotch (*Cochliobolus sativus*), leaf stripe (*Pyrenophora graminea*), and *Rhizoctonia* damping-off (*Rhizoctonia solani*). Diseases in corn: smut (*Ustilago maydis*), brown spot (*Cochliobolus heterostrophus*), copper spot (*Gloeocercospora sorghi*), southern rust (*Puccinia polysora*), gray leaf

spot (*Cercospora zeae-maydis*), white spot (*Phaeosphaeria mydis* and/or *Pantoea ananatis*) and *Rhizoctonia* damping-off (*Rhizoctonia solani*). Diseases of citrus: melanose (*Diaporthe citri*), scab (*Elsinoe fawcetti*), penicillium rot (*Penicillium digitatum*, *P. italicum*), and brown rot (*Phytophthora parasitica*, *Phytophthora citrophthora*). Diseases of apple: blossom blight (*Monilinia mali*), canker (*Valsa ceratosperma*), powdery mildew (*Podosphaera leucotricha*), *Alternaria* leaf spot (*Alternaria alternata* apple pathotype), scab (*Venturia inaequalis*), powdery mildew, bitter rot (*Colletotrichum acutatum*), crown rot (*Phytophthora cactorum*), blotch (*Diplocarpon mali*), and ring rot (*Botryosphaeria berengeriana*). Diseases of pear: scab (*Venturia nashicola*, *V. pirina*), powdery mildew, black spot (*Alternaria alternata* Japanese pear pathotype), rust (*Gymnosporangium haraeaeum*), and *phytophthora* fruit rot (*Phytophthora cactorum*). Diseases of peach: brown rot (*Monilinia fructicola*), powdery mildew, scab (*Cladosporium carpophilum*), and *phomopsis* rot (*Phomopsis* sp.). Diseases of grape: anthracnose (*Elsinoe ampelina*), ripe rot (*Glomerella cingulata*), powdery mildew (*Uncinula necator*), rust (*Phakopsora ampelopsidis*), black rot (*Guignardia bidwellii*), *botrytis*, and downy mildew (*Plasmopara viticola*). Diseases of Japanese persimmon: anthracnose (*Gloeosporium kaki*), and leaf spot (*Cercospora kaki*, *Mycosphaerella nawae*). Diseases of gourd: anthracnose (*Colletotrichum lagenarium*), powdery mildew (*Sphaerotheca fuliginea*), gummy stem blight (*Mycosphaerella melonis*), *Fusarium* wilt (*Fusarium oxysporum*), downy mildew (*Pseudoperonospora cubensis*), *Phytophthora* rot (*Phytophthora* sp.), and damping-off (*Pythium* sp.). Diseases of tomato: early blight (*Alternaria solani*), leaf mold (*Cladosporium fulvum*), and late blight (*Phytophthora infestans*). Diseases of eggplant: brown spot (*Phomopsis vexans*), and powdery mildew (*Erysiphe cichoracearum*). Diseases of cruciferous vegetables: *Alternaria* leaf spot (*Alternaria japonica*), white spot (*Cercospora brassicae*), clubroot (*Plasmodiophora brassicae*), and downy mildew (*Peronospora parasitica*). Diseases of onion: rust (*Puccinia allii*), and downy mildew (*Peronospora destructor*). Diseases of soybean: purple seed stain (*Cercospora kikuchii*), sphaceloma scab (*Elsinoe glycines*), pod and stem blight (*Diaporthe phaseolorum* var. *sojae*), *septoria* brown spot (*Septoria glycines*), frog-eye leaf spot (*Cercospora sojae*), rust (*Phakopsora pachyrhizi*), Yellow rust, brown stem rot (*Phytophthora sojae*), and *Rhizoctonia* damping-off (*Rhizoctonia solani*). Diseases of kidney bean: anthracnose (*Colletotrichum lindemthianum*). Diseases of peanut: leaf spot (*Cercospora personata*), brown leaf spot (*Cercospora arachidicola*) and southern blight (*Sclerotium rolfsii*). Diseases of garden pea: powdery mildew (*Erysiphe pisi*), and root rot (*Fusarium solani* f. sp. *pisi*). Diseases of potato: early blight (*Alternaria solani*), late blight (*Phytophthora infestans*), pink rot (*Phytophthora erythroseptica*), and powdery scab (*Spongospora subterranean* f. sp. *subterranea*). Diseases of strawberry: powdery mildew (*Sphaerotheca humuli*), and anthracnose (*Glomerella cingulata*). Diseases of tea: net blister blight (*Exobasidium reticulatum*), white scab (*Elsinoe leucospila*), gray blight (*Pestalotiopsis* sp.), and anthracnose (*Colletotrichum theae-sinensis*). Diseases of tobacco: brown spot (*Alternaria longipes*), powdery mildew (*Erysiphe cichoracearum*), anthracnose (*Colletotrichum tabacum*), downy mildew (*Peronospora tabacina*), and black shank (*Phytophthora nicotianae*). Diseases of rapeseed: *sclerotinia* rot (*Sclerotinia sclerotiorum*), and *Rhizoctonia* damping-off (*Rhizoctonia solani*). Diseases of cotton: *Rhizoctonia* damping-off (*Rhizoctonia solani*). Diseases of sugar beat: *Cercospora* leaf spot (*Cercospora beticola*), leaf blight (*Thanatephorus cucumeris*), Root rot (*Thanatephorus cucumeris*), and *Aphanomyces* root rot (*Aphanomyces cochlioides*). Diseases of rose: black spot (*Diplocarpon rosae*), powdery mildew (*Sphaerotheca pannosa*), and downy mildew (*Peronospora sparsa*). Diseases of chrysanthemum and asteraceous plants: downy mildew (*Bremia lactucae*), leaf blight (*Septoria chrysanthemi-indici*), and white rust (*Puccinia horiana*). Diseases of various groups: diseases caused by *Pythium* spp. (*Pythium aphanidermatum*, *Pythium debarianum*, *Pythium graminicola*, *Pythium irregulare*, *Pythium ultimum*), gray mold. (*Botrytis cinerea*), and *Sclerotinia* rot (*Sclerotinia sclerotiorum*). Disease of Japanese radish: *Alternaria* leaf spot (*Alternaria brassicicola*). Diseases of turfgrass: dollar spot (*Sclerotinia homeocarpa*), and brown patch and

large patch (*Rhizoctonia solani*). Disease of banana: Black sigatoka (*Mycosphaerella fijiensis*), Yellow sigatoka (*Mycosphaerella musicola*). Disease of sunflower: downy mildew (*Plasmopara halstedii*). Seed diseases or diseases in the early stages of the growth of various plants caused by *Aspergillus* spp., *Penicillium* spp., *Fusarium* spp., *Gibberella* spp., *Tricoderma* spp., *Thielaviopsis* spp., *Rhizopus* spp., *Mucor* spp., *Corticium* spp., *Phoma* spp., *Rhizoctonia* spp. and *Diplodia* spp. Viral diseases of various plants mediated by *Polymixa* spp. or *Olpidium* spp. and so on.

(77) The compositions of the present invention can be used in agricultural lands such as fields, paddy fields, lawns and orchards or in non-agricultural lands. The present invention may be used to control diseases in agricultural lands for cultivating the plants without any phytotoxicity to the plant.

(78) Examples of the crops on which the present compositions may be used include but are not limited to corn, rice, wheat, barley, rye, oat, sorghum, cotton, soybean, peanut, buckwheat, beet, rapeseed, sunflower, sugar cane, tobacco, etc.; vegetables: solanaceous vegetables such as eggplant, tomato, pimento, pepper, potato, etc., cucurbit vegetables such as cucumber, pumpkin, zucchini, water melon, melon, squash, etc., cruciferous vegetables such as radish, white turnip, horseradish, kohlrabi, Chinese cabbage, cabbage, leaf mustard, broccoli, cauliflower, etc., asteraceous vegetables such as burdock, crown daisy, artichoke, lettuce, etc, liliaceous vegetables such as green onion, onion, garlic, and asparagus, ammiaceous vegetables such as carrot, parsley, celery, parsnip, etc., chenopodiaceous vegetables such as spinach, Swiss chard, etc., lamiaceous vegetables such as *Perilla frutescens*, mint, basil, etc, strawberry, sweet potato, *Dioscorea japonica*, colocasia, etc., flowers, foliage plants, turf grasses, fruits: pome fruits such apple, pear, quince, etc, stone fleshy fruits such as peach, plum, nectarine, *Prunus mume*, cherry fruit, apricot, prune, etc., citrus fruits such as orange, lemon, lime, grapefruit, etc., nuts such as chestnuts, walnuts, hazelnuts, almond, pistachio, cashew nuts, macadamia nuts, etc. berries such as blueberry, cranberry, blackberry, raspberry, etc., grape, kaki fruit, olive, plum, banana, coffee, date palm, coconuts, etc., trees other than fruit trees; tea, mulberry, flowering plant, trees such as ash, birch, dogwood, Eucalyptus, Ginkgo biloba, lilac, maple, Quercus, poplar, Judas tree, Liquidambar formosana, plane tree, zelkova, Japanese arborvitae, fir wood, hemlock, juniper, Pinus, Picea, and Taxus cuspidate, etc.

(79) In an embodiment, the constituent fungicides of the combination of the present invention may be admixed in ratio of (1-80):(1-80):(1-80):(1:80)

(80) In an aspect, the present invention may provide methods of controlling fungal diseases comprising applying a combination comprising: (a) at least one succinate dehydrogenase inhibitor fungicide; (b) at least one quinone outside inhibitor and/or at least one ergosterol biosynthesis inhibitor; and (c) at least one dithiocarbamate fungicide.

(81) In an embodiment, the succinate dehydrogenase inhibitor fungicide, the quinone outside inhibitor fungicide, the ergosterol biosynthesis inhibitor fungicide, and the dithiocarbamate fungicide may be selected according to any of the preferred embodiments of the combinations described hereinabove.

(82) The combinations of the present invention may be sold as a pre-mix composition or a kit of parts such that individual actives may be mixed before spraying. Alternatively, the kit of parts may contain succinate dehydrogenase inhibitor fungicide and the dithiocarbamate fungicide pre-mixed and the third active may be admixed with an adjuvant such that the two components may be tank mixed before spraying.

(83) The composition of the present invention may be applied simultaneously as a tank mix or a formulation or may be applied sequentially. The application may be made to the soil before emergence of the plants, either pre-planting or post-planting. The application may be made as a foliar spray at different timings during crop development, with either one or two applications early or late post-emergence.

(84) The compositions according to the invention can be applied before or after infection of the useful plants or the propagation material thereof by the fungi.

(85) As will be demonstrated in the examples, the addition of a dithiocarbamate fungicide to a combination of succinate dehydrogenase inhibitors which are combined with Quinone outside inhibitors and/or ergosterol biosynthesis inhibitors, greatly improved the disease control as well as improved yield and demonstrated a synergistic effect. The lower the mixture performance in the disease control, the greater the additional benefit of the mancozeb when added to the compositions of the present invention.

(86) Examples: Studies were conducted to study the addition of dithiocarbamate fungicides to succinate dehydrogenase inhibitor fungicides and at least one other fungicide and the contribution of the dithiocarbamate to the efficacy of these mixtures. Experiments were conducted over a period of two years to study the effect of the addition of dithiocarbamates on the efficacy of succinate dehydrogenase inhibitors alone and when combined with a co-fungicide such as a Quinone outside inhibitors and/or ergosterol biosynthesis inhibitor fungicide. Doses tested were at rates of 1500 g/ha for Mancozeb, 150 ml/ha for the ergosterol biosynthesis inhibitor, 200 g/ha for the Quinone outside inhibitor, 1000 g/ha and 200 g/ha for succinate dehydrogenase inhibitor fungicides. The combinations were tested for efficacy of disease control in soybean for the control of Asian soybean rust. The trials were carried out at various locations in India.

(87) The tests were conducted on soybean cultivar Monsoy 9144 RR. The commercially available individual active ingredients were used at the indicated dosages.

(88) TABLE-US-00011 TABLE 1 Table 1 shows the efficacy of mancozeb when added to combinations containing succinate dehydrogenase inhibitor fungicides and ergosterol biosynthesis inhibitor. Mean percent disease control Treatment Dose rates (ml/g/ha) 2015/16 2016/17
Prothioconazole + 150 + 200 88.04 71.04 Benzovindiflupyr Prothioconazole + 150 + 200 + 1500 95.07 95.07 Benzovindiflupyr + Mancozeb Prothioconazole + Isopyrazam 150 + 1000 87.14 70.23
Prothioconazole + Isopyrazam + 150 + 1000 + 1500 93.4 93.07 Mancozeb Prothioconazole + Penthiopyrad 150 + 1000 83.33 68.15 Prothioconazole + 150 + 1000 + 1500 93.07 92.18
Penthiopyrad + Mancozeb

Table 1 clearly demonstrates the increase in control when mancozeb is added to the combination of ergosterol biosynthesis inhibitors and succinate dehydrogenase inhibitor fungicides.

(89) TABLE-US-00012 TABLE 2 Table 2 demonstrates efficacy when mancozeb is added to combinations of succinate dehydrogenase inhibitor fungicides and quinone outside inhibitor fungicides and ergosterol biosynthesis inhibitors. Mean percent disease control Treatment Dose rates (ml/g/ha) 2015/16 2016/17 Prothioconazole + 150 + 200 + 500 93.07 91.07 Benzovindiflupyr + Azoxystrobin Prothioconazole + 150 + 200 + 500 + 1500 96.36 95.03 Benzovindiflupyr + Azoxystrobin + Mancozeb

Table 2 clearly demonstrates the importance of adding mancozeb to the combination. The addition of mancozeb improved disease control.

(90) It was thus found that the incorporation of mancozeb greatly increased efficacy and disease control of the succinate dehydrogenase inhibitor fungicide treatments when mixed with ergosterol biosynthesis inhibitors or quinone outside inhibitors. It was thus concluded that the addition of mancozeb enhanced the efficacy of the combination and surprisingly gave a synergistic effect. The addition of a dithiocarbamate increased disease control and improved yield of plants. The instant invention is more specifically explained by above examples. However, it should be understood that the scope of the present invention is not limited by the examples in any manner. It will be appreciated by any person skilled in this art that the present invention includes aforesaid examples and further can be modified and altered within the technical scope of the present invention.

(91) While the foregoing written description of the invention enables one of ordinary skill to make and use what is considered presently to be the best mode thereof, those of ordinary skill will understand and appreciate the existence of variations, combinations, and equivalents of the specific embodiment, method, and examples herein. The invention should therefore not be limited by the

above described embodiment, method, and examples, but by all embodiments and methods within the scope and spirit of the invention.

Claims

1. A fungicidal combination consisting of: mancozeb, prothioconazole, and benzovindiflupyr.
 2. The combination as claimed in claim 1, wherein the combination is formulated into wettable powders, granules, dusts, soluble concentrates, suspension concentrates, oil in water emulsion, water in oil emulsion, emulsifiable concentrates, capsule suspensions, suspension concentrates combined with capsule suspensions, or oil dispersions.
 3. A method of foliar application comprising applying the combination of claim 1 to a plant propagation material.
 4. A method of controlling fungal diseases in plants comprising applying to a locus of the plant the combination of claim 1.
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